

Midterm N°2

Name : _____ Groupe: _____

Exercise 1. Among these assertions, specify which ones are true and which ones are false.

	True	False
1- If $P(A \cup B) = 0.5$ and $P(A \cap B) = 0.5$, then $P(A) = P(B)$.	✓	
2- if $P(A) = 0.4$, $P(B) = 0.5$ and $P(A \cup B) = 0.7$ then $P(A \cap B) = 0.3$.		✓
3- If $P(A B) = 0.5$ and $P(B) = 0.5$, then A and B are necessarily independent.		✓
4- If two events are independent, then $p(A B) = p(B A)$.		✓
5- If two events are independent then both $P(A B) = P(A)$ and $P(B A) = P(A)$.		✓
6- If $P(A B) = 0.6$ and $P(B) = 0.2$ then $P(A \cap B) = 0.12$.	✓	
7- If $P(A \cap B) \geq 0.10$ then $P(A) \geq 0.10$.	✓	
8- If A and B are mutually exclusive events, then $P(A B) = 0$.	✓	
9- Bayes' rule states that $p(A B) = p(B A)$.		✓
10- A certainty event has a probability equal to 0.9.		✓
11- Any random variable is a measurable function.	✓	
12- A random variable can be qualitatif		✓
13- $X \stackrel{d}{=} Y$ if $\forall x \in \mathbb{R}, F_X(x) = F_Y(x)$.	✓	
14- $x_1 \leq x_2 \implies F(x_1) \leq F(x_2)$.	✓	
15- B and $A \cap B^c$ are disjoint.	✓	
16- $P(A) - P(B) \neq P(A \cap B^c)$.		✓
17- If $A_i =]0, \frac{1}{i}[$ then $\lim_{i \rightarrow \infty} P(A_i) = 0$.	✓	
18- $(A \cap B)^c = A^c \cup B^c$ and $(A \cup B)^c = A^c \cap B^c$.		✓
19- $P(A_1 \cup A_2 \cup \dots) \geq P(A_1) + P(A_2) + \dots$		✓
20- $E[XY] = E[X]E[Y]$.		✓

Exercise 2. We give the following table

x_i	1	2	3	4	5	6	7	8	9	10
y_i	1504	809	540	438	375	320	290	254	231	212
$(x_i - \bar{X})$	-4.5	-3.5	-2.5	-1.5	-0.5	0.5	1.5	2.5	3.5	4.5
$(y_i - \bar{Y})$	1006.7	311.7	42.7	-59.3	-122.3	-177.3	-207.3	-243.3	-266.3	-285.3
$(x_i - \bar{X})(y_i - \bar{Y})$	-4530.15	-1090.95	-106.75	88.95	61.15	-88.65	-310.95	-608.25	-932.05	-1283.85
R_{y_i}	10	9	8	7	6	5	4	3	2	1
d_i^2	81	49	25	9	1	1	9	25	49	81

1. Calculate the correlation coefficients of pearson and spearman then explain the difference between the two obtained results

$$\begin{aligned} \bar{X} &= 5.5 \quad , \quad \bar{Y} = 497.3 \\ \sum (x_i - \bar{X})^2 &= 82.5 \quad , \quad \sum (y_i - \bar{Y})^2 = 1416814.1 \quad , \quad \sum (x_i - \bar{X})(y_i - \bar{Y}) = -8801.5 \\ \sum d_i^2 &= 330 \end{aligned}$$

We have

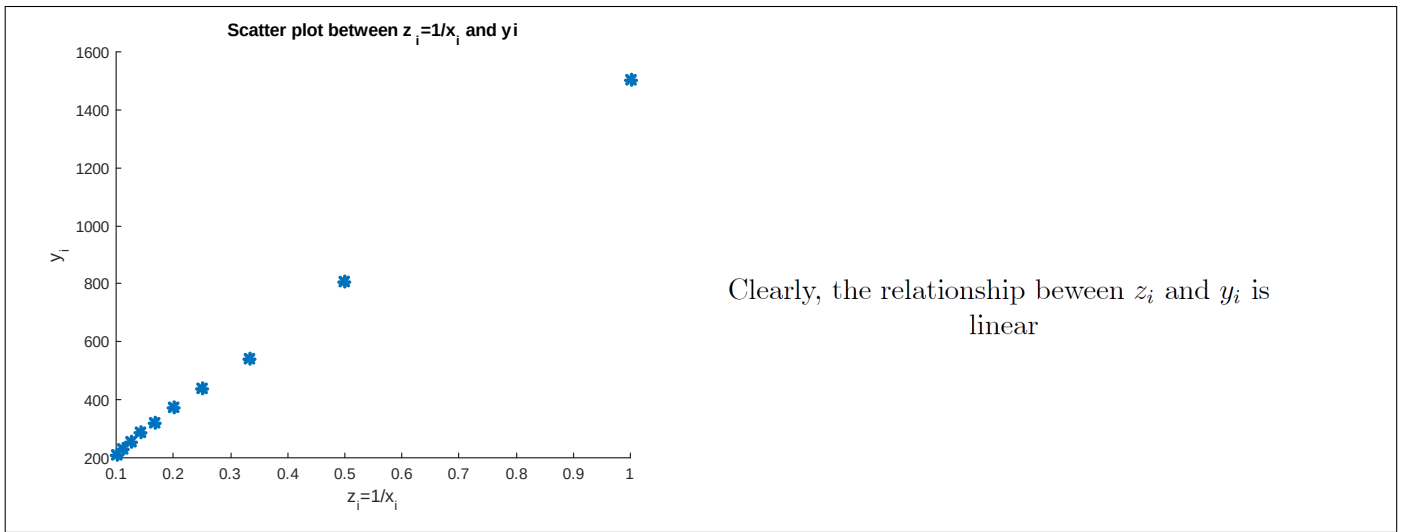
$$\rho(X, Y) = \frac{\sum (x_i - \bar{X})(y_i - \bar{Y})}{\sqrt{\sum (x_i - \bar{X})^2 \sum (y_i - \bar{Y})^2}} = \frac{-8801.5}{\sqrt{82.5 \times 1416814.1}} = -0.81409$$

and

$$r_s = 1 - \frac{6 \sum d_i^2}{N(N^2 - 1)} = 1 - \frac{6 \times 330}{10 \times (10^2 - 1)} = -1.0$$

Spearman's coefficient is equal to -1, which indicates a perfect negative monotonic trend, and since Pearson's coefficient is not equal to r_s , which means that the relationship is not perfect negative linear, therefore, the relationship between the number of units and the unit price is not linear.

2. Set $z_i = \frac{1}{x_i}$, calculate z_i then draw the scatter plot between z_i and the unit price y_i . Comment.



Exercise 3. Consider the data in the table below. In this table Y is the purity of oxygen produced in a chemical distillation process, and X is the percentage of hydrocarbons that are present in the main condenser of the distillation unit.

Obs. N°	X	Y	$(x_i - \bar{X})$	$(y_i - \bar{Y})$	$(x_i - \bar{X})(y_i - \bar{Y})$	$y_i - \hat{y}_i$	$\hat{y}_i - \bar{Y}$
1	0.99	90.01	-0.194	-2.247	0.436	0.652	-2.899
2	1.02	89.05	-0.164	-3.207	0.526	-0.756	-2.451
3	1.15	91.43	-0.034	-0.827	0.028	-0.319	-0.508
4	1.29	93.74	0.106	1.483	0.157	-0.101	1.584
5	1.46	96.73	0.276	4.473	1.235	0.349	4.125
6	1.36	94.45	0.176	2.193	0.386	-0.437	2.630
7	0.87	87.59	-0.314	-4.667	1.465	0.026	-4.692
8	1.23	91.77	0.046	-0.487	-0.022	-1.174	0.687
9	1.55	99.42	0.366	7.163	2.622	1.694	5.470
10	1.40	93.65	0.216	1.393	0.301	-1.835	3.228
11	1.19	93.54	0.006	1.283	0.008	1.194	0.090
12	1.15	92.52	-0.034	0.263	-0.009	0.771	-0.508
13	0.98	90.56	-0.204	-1.697	0.346	1.352	-3.049
14	1.01	89.54	-0.174	-2.717	0.473	-0.116	-2.600
15	1.11	89.85	-0.074	-2.407	0.178	-1.301	-1.106

1. Measure the correlation between the two variables then interpret it.

$$\bar{X} = 1.184 \quad , \quad \bar{Y} = 92.257$$

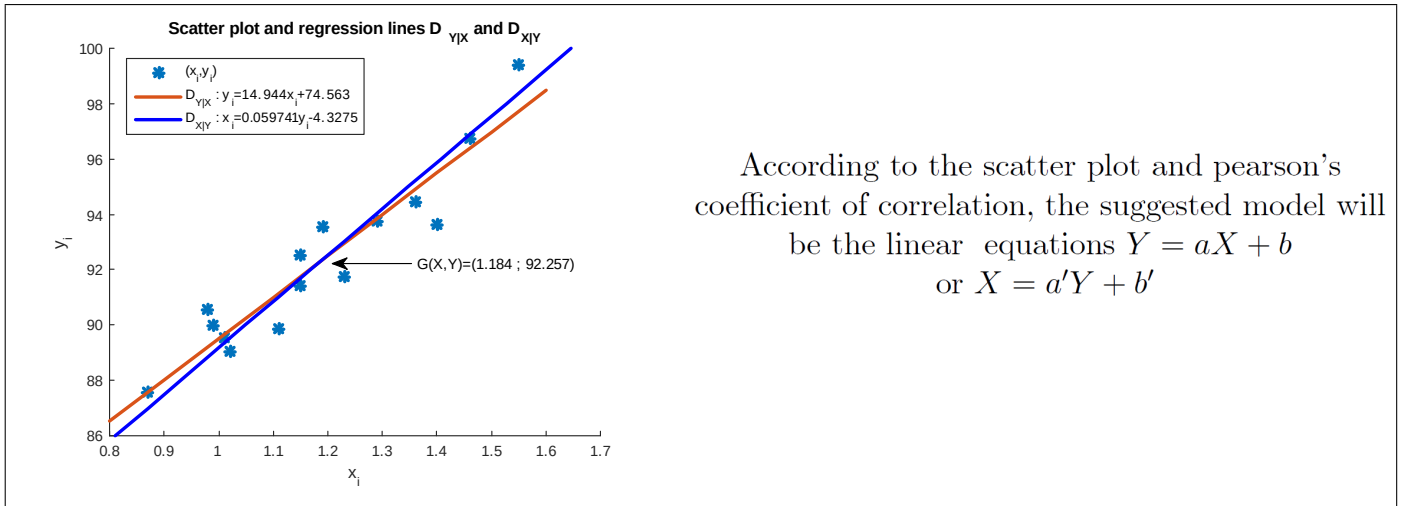
$$\sum (x_i - \bar{X})^2 = 0.54396 \quad , \quad \sum (y_i - \bar{Y})^2 = 136.0719333 \quad , \quad \sum (x_i - \bar{X})(y_i - \bar{Y}) = 8.1291$$

Pearson's coefficient of correlation is given by

$$\rho(X, Y) = \frac{\sum (x_i - \bar{X})(y_i - \bar{Y})}{\sqrt{\sum (x_i - \bar{X})^2 \sum (y_i - \bar{Y})^2}} = \frac{8.1291}{\sqrt{0.54396 \times 136.0719333}} = 0.94488$$

which indicate a very strong positive linear correlation between the percentage of hydrocarbons and the purity of oxygen produced.

2. Draw the scatter plot between the two variables. Which model is the most appropriate to fit our data.



According to the scatter plot and pearson's coefficient of correlation, the suggested model will be the linear equations $Y = aX + b$ or $X = a'Y + b'$

3. We aim to explain in both sens the relationship between Y and X . Let

$$Y = aX + b + \varepsilon \quad \text{and} \quad X = a'Y + b' + \varepsilon'.$$

Give the values of the parameters a, a', b and b' for which $\sum \varepsilon^2$ and $\sum (\varepsilon')^2$ are minimals.

$$a = \frac{\sum (x_i - \bar{X})(y_i - \bar{Y})}{\sum (x_i - \bar{X})^2} = \frac{8.1291}{0.54396} = 14.944$$

and

$$b = \bar{Y} - a\bar{X} = 92.257 - 14.944 \times 1.184 = 74.563.$$

Thus, the regression line $D_{Y|X}$ will be $y_i = 14.944x_i + 74.563$. Also

$$a' = \frac{\sum (x_i - \bar{X})(y_i - \bar{Y})}{\sum (y_i - \bar{Y})^2} = \frac{8.1291}{136.0719333} = 0.059741$$

and

$$b' = \bar{X} - a'\bar{Y} = 1.184 - 0.059741 \times 92.257 = -4.3275.$$

The regression line $D_{X|Y}$ will be $x_i = 0.059741y_i - 4.3275$.

4. Plot in the above figure the two lines $D_{Y|X}$ and $D_{Y|X}$ represented by the two equations above. What are the coordinates of the point of intersection between the two lines and what does it represent?

the intersection point between the two lines $D_{Y|X}$ and $D_{Y|X}$ is $G(\bar{X}, \bar{Y}) = (1.184; 92.257)$ and it represents the center of the cloud of the scatter plot also known as the average/center of gravity

5. Compute the residual variance and the variance explained by the regression line $D_{Y|X}$. Do you think that the regression line fit well the data, measure it.

We call residual variance of Y or residual variance of $D_{Y|X}$, and we note $V_R(Y)$, the following expression

$$V_R(Y) = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 = 0.97248$$

We call variance explained by the regression line $D_{Y|X}$, and we note $V_E(Y)$, the expression defined by

$$V_E(Y) = \frac{1}{n} \sum_{i=1}^n (\hat{y}_i - \bar{Y})^2 = 8.099.$$

We know that $Var(Y) = V_R(Y) + V_E(Y)$ which gives that

$$V_E(Y) = \frac{Cov(X, Y)^2}{Var(X)} = Var(Y) \rho(X, Y)^2 = 9.071462 \times 0.94488^2 = 8.099$$

$$V_R(Y) = Var(Y)(1 - \rho(X, Y)^2) = 9.071462 \times (1 - 0.94488^2) = 0.97248.$$

Both regression lines fit well the cloud of points in the above plot, to measure the goodness of fit, we need to compute the coefficient of determination

$$R^2 = \frac{V_E(Y)}{Var(Y)} = \frac{8.099}{9.071462} = 0.893$$

which is close to 1.

6. According to you, what is the level of hydrocarbon required to achieve 100% oxygen purity and what is the least percentage of oxygen purity at a hydrocarbon level of 0.8.

To achieve 100% oxygen purity, the estimated level of hydrocarbon required is given by $D_{X|Y} : x_{100\%} = 0.059741 \times 100 - 4.3275 = 1.6466$ and at a hydrocarbon level of 0.8, the minimum percentage of oxygen purity is obtained by $D_{Y|X} : y_{0.8} = 14.944 \times 0.8 + 74.563 = 86.518\%$.